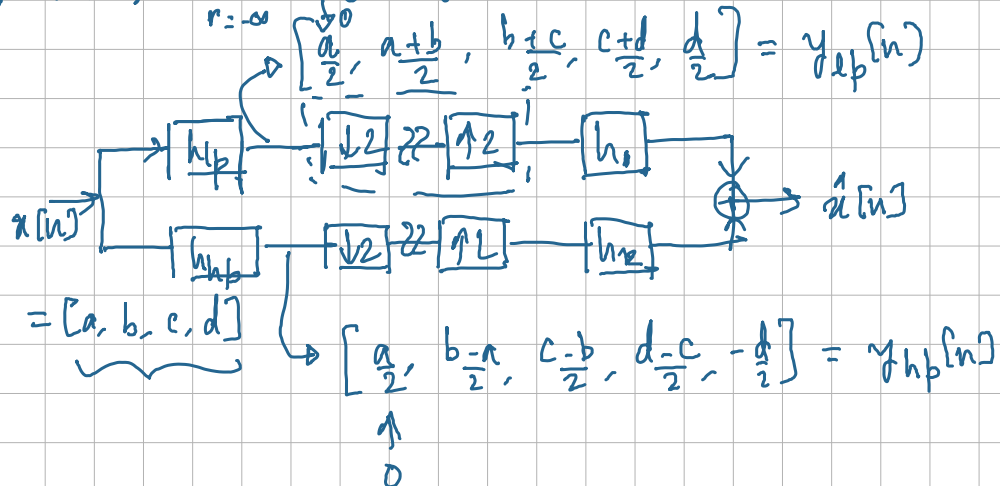


17/09/26

Signal Processing

- Recap: DFT properties and theorems; $x[n] = a^n \cdot u[n]$; $|a| < 1$; Q4(b)
- Useful theorems: Linearity, time delay*, modulation*, time flip, convolution*, Parseval's theorem
- Periodicity; $X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi r)$

Q4b



$$4(a). y_{lp}[n] = \left[\frac{a}{2}, \frac{a+b}{2}, \frac{b+c}{2}, \frac{c+d}{2}, \frac{d}{2} \right]$$

$$y_{hp}[n] = \left[\frac{a}{2}, \frac{a-b}{2}, \frac{c-b}{2}, \frac{d-c}{2}, -\frac{d}{2} \right]$$

$$4(b) \text{ output at compressor: } \frac{a}{2}, \frac{b+c}{2}, \frac{d}{2}$$

$$\frac{a}{2}, \frac{b-c}{2}, -\frac{d}{2}$$

$$\text{output at upsampler: } \frac{a}{2}, 0, \frac{b+c}{2}, 0, \frac{d}{2}$$

$$\frac{a}{2}, 0, \frac{c-b}{2}, 0, -\frac{d}{2}$$

$$\bullet h_1[n] = 2h_{lp}[n]$$

$$h_2[n] = -2h_{hp}[n]$$

$$\bullet \text{ output at } h_1[n] = \left[\left(\frac{a+0}{2} \right), \left(\frac{a}{2} \right), \left(\frac{b+c+0}{2} \right), \dots \right]$$

$$\text{output at } h_2[n] = \left[-\left(\frac{a}{2} \right), \left(0 + \frac{a}{2} \right), \left(\frac{b-c}{2} \right), \dots \right]$$

$$= [0, a, b, c, d, \dots, 0, \dots]$$

Several observations:

- This is a perfect reconstruction filter bank, i.e., we can exactly recover $x[n]$ from y_{lp} y_{hp} .
- This is a Haar wavelet implementation, a very useful signal analysis tool.
- $\downarrow 2$ helps maintain signal size
- The decomposition into y_{lp} , y_{hp} has several useful applications including communication, storage etc.